Gonzalo M. Díaz

Meteorologist (M. Sc.)
Data Scientist

20 oct 1985

Buenos Aires, Argentina

2. Education

08/2006 - 12/2015

Master's Degree in Atmospheric Sciences (M.Sc.)

Department of Atmospheric and Ocean Sciences. Facultad de Ciencias Exactas y Naturales
University of Buenos Aires

Master's thesis: "Adjustment of a one-dimensional soil moisture model in localities of the province of Entre Ríos using in-situ observations and remote sensing".

January 2012

English (B2)
First Certificate in English (FCE)

University of Cambridge

Reading	8 / 10
Talking	6 / 10
Writing	7 / 10

03/1999 - 12/2004

Electronic and Electromechanical Technician

Technological Institute Philips Argentina

1. Work experience

Scientific Application Analyst with a focus on Climate Services for Water Resources | National Meteorological Service (NMS) — **12/2012 - today**

Department of Science, Technology and Production
Ministry of Defence

Update and development of new products and services for the NMS website and for the different users belonging to the different productive sectors of the country (Water Resources, Agriculture, Energy, etc.). Development of new statistical and Machine Learning tools for the development of predictive models and for use in user interface web applications. Development of web platforms for control and query of meteorological data from the stations of the NMS and third parties. Implementation of operational hydrological models for hydrometric heights and flood monitoring. Preparation of damage reports, due to meteorological phenomena, intended to private companies and other areas of the NMS and by legal requirements.

Head of Practical Work (head teacher) — **03/2021 - today**

Subject: Statistical Methods and Hydrology
Department of Atmospheric and Ocean Sciences
Facultad de Ciencias Exactas y Naturales. University of Buenos Aires (UBA)

Part-time research consultant — **01/2018 - 12/2025**

Department of Atmospheric and Ocean Sciences
Facultad de Ciencias Exactas y Naturales. University of Buenos Aires (UBA)

UBACyT Project 20020170100117BA
Daily extreme precipitation events in southeastern South America: statistical and dynamic climate modeling and evaluation of its hydrological impacts.

PICT-2018-02496 Project
Statistical and dynamic modeling of daily extreme precipitation events in southeastern South America.

Professor — **03/2020 - today**

Master in Water Management.
Subject: Climate System
Facultad de Ciencias Veterinarias. University of Buenos Aires (UBA)

3. Most relevant publications

- **Díaz G.**, & Ferreira L. (2025). "Development of web applications in meteorology for the agricultural sector". In: *Argentine Conference on Informatics and Operational Research (54JAIIO). Argentine Congress of Agroinformatics (CAI), Argentina, Aug 04-07*.
- **Díaz G.**, Chiavetta A., Siches J., Santoro F., Rueda D., Moisés S., Cappa C., González Lebrero C., Maffey L., Fattore G., Utgés M.E. Manteca M. & Ferreira L. (2025). "Application of Machine Learning Models Using Epidemiological and Meteorological Variables for the Prediction of Dengue Cases". En: *15th Argentine Congress of Meteorology (XV CONGREMET), Argentina, Sep 30-Oct 03.*
- Doyle M., **Díaz G.**, & Chavez L. (2024). "Ability of downscaling methods to represent extreme precipitation events and the impact on the Uruguay River streamflow". In: *GeoActa Vol 46 Núm 1.*

4. Barskills

R language / Shiny library 80%

Own R libraries creation
CRAN link: aws.wrfsmn
GitHub link: era5.aws 60%

Linux environment 70%

MATLAB / Octave 70%

GrADS language 65%

Python / APIs devs 40%

GitHub 40%

Q-GIS software 60%

5. Distinctions

- **Distinction for the best work team in the Fifth Workshop on Water Resources in Developing Countries**. *(Collective award)*.
  Name of the work: A performance comparison between the CHyM and VIC models for a subbasin within La Plata Basin.
  Given by International Centre for Theoretical Physics (ICTP)
- **Diploma of honor for "Meritorious Performance for Joint Work"**. *(Collective award)*.
  Given by the National Meteorological Service

7. Contact

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City of Buenos Aires

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github.com/Gonzalo1985

- **Díaz G.**, Alvarez Imaz M., Bonel N., González Morinigo C., Marcora G. & Bontempi E. (2024). "Co-design of Temperature and Humidity Index forecast products for the agricultural sector". In: *Technical Note SMN 2024-182*.
- **Díaz G.**, Cerrudo C., Godoy A., & Ferreira L. (2024). "Value of hydroclimatic information for decision making in the hydropower sector from a soil model: Correction with multiple linear regressions". In: *XXX scientific meeting of the Argentine association of Geophysicists and Geodesists, Argentina, Apr 15-19*.
- Cerrudo C., Godoy A., **Díaz G.**, Garbarini E. & Ferreira L. (2024). "A climate service co – production experience tailored to the hydroenergy sector in South America". In: *2nd WMO/WWRP Weather & Society Conference*.
- Hobouchian P., **Díaz G.**, Vidal L., García Skabar Y., de Elía R., Ferreira L., Maas M., Rossi Lopardo M.S., Veiga H. and Rugna M. (2021). "Adjustment of the IMERG satellite precipitation estimation with pluviometric observations in Argentina". In *Technical Note SMN 2021-105*.
- Doyle M., Chavez L. & **Díaz, G.**,(2021). "Hydrological simulations of the Uruguay river based on observational and hybrid precipitation data sources". In: *XXIX scientific meeting of the Argentine association of Geophysicists and Geodesists, Argentina, Aug 02-10*.
- **Díaz G.**, Vita M., Hobouchian P., Ferreira L. and Giordano L. (2021). "Expansion of the NMS network using precipitation data from third-party automatic weather stations". In *Technical Note SMN 2021-90*.
- **Díaz, G.**, Doyle M. & Chavez L. (2020). "Impact of satellite precipitation estimations on the flows of the Uruguay River". In: *15th International Meeting of Earth Sciences, Mendoza, Argentina, Nov 23-25*.
- **Díaz, G.**, Oliveri P., Meis M. & Bianchi E. (2019). "Preliminary analysis of the performance of the CHyM and VIC hydrological models for the Uruguay river basin". In: *IV National Congress of Environmental and Technology Science, Buenos Aires, Argentina, Dec 02-05*.
- Solman S., Bettolli M.L., Doyle M., Blazquez J., Feijoo M., Olmo M., **Díaz, G.**, Balmaceda R. & Poggi M. (2019). "How well do RCMs and ESDs reproduce the occurrence of extreme precipitation events over Southeastern South America? A case study approach". In: *International Conference on Regional Climate CORDEX 2019. Beijing, China, Oct 14-18*.
- Cerrudo, C., **Díaz, G.**, Juárez S. & Ferreira L. (2017). "Analysis of the spatial-temporal relationship between the precipitation estimated by the TRMM satellite (3B42RT) and the average daily flow in the Iguazu river basin". In: *Meteorológica*.
- **Díaz, G.**, Doyle M. & Brizuela A. (2014). "Calibration of a one-dimensional hydrological model for the Diamante and Parana stations, Entre Rios". In: *Meteorológica*.

6. Training

- Forwards Taskforce Package Development Workshops
  (2026) May to July (virtual). Provided by: R Forwards. *Coordinators: Joyce Robbins, Jesi Formoso and Heather Turner*. Duration time: 10 hours.
- Using the (g)EMOS methodology for South America and machine learning (ML) examples at MeteoSwiss
  (2024) November 12 to 14 (virtual). Provided by: MeteoSwiss and Regional Center Formation WMO-SMN. *Coordinators: area experts*. Duration time: 12 hours.
- Development of subseasonal prediction tools for southern South America
  (2024) July 04 to August 01 (virtual). Provided by: SISSA and Regional Center Formation WMO-SMN. *Coordinators: area experts*. Duration time: 12 hours.
- KCCP: Reinforcement of Meteorological Services
  (2023) October 01 to December 13. Tokyo, Japan. Provided by: Japan International Cooperation Agency (JICA) and Japanese Meteorological Agency (JMA). *Coordinators: area experts*. Duration time: 440 hours.

- Satellite data calibration methodology with Machine Learning
  (2023) May 15 to 18 (virtual) and July 24 to 27. Buenos Aires, Argentina. Provided by: WMO SMN Regional Training Center. *Coordinators: area experts*.

- Course of Machine Learning
  (2020) During 2020. Stanford, United States of America (virtually). Provided by: Stanford University, through Coursera platform. *Coordinator: Andrew Ng and others*.

- Global Flash Flood Guidance System Workshop
  (2019) November 04 to 08. Antalya, Turkey. Provided by: Hydrologic Research Center (HRC), National Weather Service of the U.S. National Oceanic and Atmospheric Administration (NOAA). *Coordinators: Konstantine Georgakakos and others* Duration time: 40 hours.

- Fifth Workshop on Water Resources in Developing Countries: Hydroclimate Modeling and Analysis Tools
  (2019) May 27 to June 07. Trieste, Italy. Provided by: International Centre for Theoretical Physics (ICTP). *Coordinators: Soroosh Sorooshian, Marco Verdecchia, Erika Coppola and Fabio di Sante.* Duration time: 74 hours.

- Applied statistics in meteorology with R language
  (2015) August 27 to September 01. Buenos Aires, Argentina. *Coordinator: Dr. Gustavo Naumann.* Duration time: 20 hours.

Buenos Aires, 3rd June 2026